

1- Etcd

```
root@controlplane:~# ps -ef | grep etcd
root      1750   1597   2 11:37 ?        00:01:38 kube-apiserver --advertise-address=172.30.1.2 --allow-privileged=true --authorization-mode=Node,RBAC --client-ca-file=/etc/kubernetes/pki/ca.crt --enable-admission-plugins=NodeRestriction --enable-bootstrap-token-auth=true --etcd-cafile=/etc/kubernetes/pki/etcd/ca.crt --etcd-certfile=/etc/kubernetes/pki/apiserver-etcd-client.crt --etcd-keyfile=/etc/kubernetes/pki/apiserver-etcd-client.key --etcd-servers=https://127.0.0.1:2379 --kubelet-client-certificate=/etc/kubernetes/pki/apiserver-kubelet-client.crt --kubelet-client-key=/etc/kubernetes/pki/apiserver-kubelet-client.key --kubelet-preferred-address-types=InternalIP,ExternalIP,Hostname --proxy-client-cert-file=/etc/kubernetes/pki/front-proxy-client.crt --proxy-client-key-file=/etc/kubernetes/pki/front-proxy-client.key --requestheader-allowed-names=front-proxy-client --requestheader-client-ca-file=/etc/kubernetes/pki/front-proxy-ca.crt --requestheader-extra-headers-prefix=X-Remote-Extra- --requestheader-group-headers=X-Remote-Group --requestheader-username-headers=X-Remote-User --secure-port=6443 --service-account-issuer=https://kubernetes.default.svc.cluster.local --service-account-key-file=/etc/kubernetes/pki/sa.pub --service-account-signing-key-file=/etc/kubernetes/pki/sa.key --service-cluster-ip-range=10.96.0.0/12 --tls-cert-file=/etc/kubernetes/pki/apiserver.crt --tls-private-key-file=/etc/kubernetes/pki/apiserver.key
root      1795   1581   1 11:37 ?        00:00:47 etcd --advertise-client-urls=https://172.30.1.2:2379 --cert-file=/etc/kubernetes/pki/etcd/server.crt --client-cert-auth=true --data-dir=/var/lib/etcd --feature-gates=InitialCorruptCheck=true --initial-advertise-peer-urls=https://172.30.1.2:2380 --initial-cluster=controlplane=https://172.30.1.2:2380 --key-file=/etc/kubernetes/pki/etcd/server.key --listen-client-urls=https://127.0.0.1:2379,https://172.30.1.2:2379 --listen-metrics-urls=http://127.0.0.1:2381 --listen-peer-urls=https://172.30.1.2:2380 --name=controlplane --peer-cert-file=/etc/kubernetes/pki/etcd/peer.crt --peer-client-cert-auth=true --peer-key-file=/etc/kubernetes/pki/etcd/peer.key --peer-trusted-ca-file=/etc/kubernetes/pki/etcd/ca.crt --snapshot-count=10000 --trusted-ca-file=/etc/kubernetes/pki/etcd/ca.crt --watch-progress-notify-interval=5s
root      4368   4240   0 12:32 pts/0    00:00:00 grep --color=auto etcd

root@controlplane:~# crictl ps | grep -E "NAME|etcd"
CONTAINER    IMAGE                CREATED          STATE      NAME      ATTEMPT     POD ID          POD          NAMESPACE
a2085e38a1cd7 0a108f7189562       55 minutes ago  Running   etcd      2           b0bfc80f23bb2  etcd-controlplane  kube-system

root@controlplane:~# cat /etc/kubernetes/manifests/etcd.yaml
apiVersion: v1
kind: Pod
metadata:
  annotations:
    kubernetes.io/etcd.advertise-client-urls: https://172.30.1.2:2379
  labels:
    component: etcd
    tier: control-plane
  name: etcd
  namespace: kube-system
spec:
  containers:
  - command:
    - etcd
    - --advertise-client-urls=https://172.30.1.2:2379
    - --cert-file=/etc/kubernetes/pki/etcd/server.crt
    - --client-cert-auth=true
    - --data-dir=/var/lib/etcd
    - --feature-gates=InitialCorruptCheck=true
    - --initial-advertise-peer-urls=https://172.30.1.2:2380
    - --initial-cluster=controlplane=https://172.30.1.2:2380
    - --key-file=/etc/kubernetes/pki/etcd/server.key
    - --listen-client-urls=https://127.0.0.1:2379,https://172.30.1.2:2379
    - --listen-metrics-urls=http://127.0.0.1:2381
    - --listen-peer-urls=https://172.30.1.2:2380
    - --name=controlplane
    - --peer-cert-file=/etc/kubernetes/pki/etcd/peer.crt
    - --peer-client-cert-auth=true
    - --peer-key-file=/etc/kubernetes/pki/etcd/peer.key
    - --peer-trusted-ca-file=/etc/kubernetes/pki/etcd/ca.crt
    - --snapshot-count=10000
    - --trusted-ca-file=/etc/kubernetes/pki/etcd/ca.crt
    - --watch-progress-notify-interval=5s
    image: registry.k8s.io/etcd:3.6.6-0
    imagePullPolicy: IfNotPresent
    livenessProbe:
      failureThreshold: 8
      httpGet:
        host: 127.0.0.1
        path: /livez
        port: probe-port
        scheme: HTTP
      initialDelaySeconds: 10
      periodSeconds: 10
      timeoutSeconds: 15
    name: etcd
    ports:
    - containerPort: 2381
      name: probe-port
      protocol: TCP
    readinessProbe:
      failureThreshold: 3
      httpGet:
        host: 127.0.0.1
        path: /readyz
        port: probe-port
        scheme: HTTP
      periodSeconds: 1
      timeoutSeconds: 15
    resources:
      requests:
        cpu: 25m
        memory: 100Mi
    startupProbe:
      failureThreshold: 24
      httpGet:
        host: 127.0.0.1
        path: /readyz
        port: probe-port
        scheme: HTTP
      initialDelaySeconds: 10
      periodSeconds: 10
      timeoutSeconds: 15
    volumeMounts:
    - mountPath: /var/lib/etcd
      name: etcd-data
    - mountPath: /etc/kubernetes/pki/etcd
      name: etcd-certs
  hostNetwork: true
  priority: 2000001000
  priorityClassName: system-node-critical
  securityContext:
    seccompProfile:
      type: RuntimeDefault
  volumes:
  - hostPath:
      path: /etc/kubernetes/pki/etcd
      type: DirectoryOrCreate
    name: etcd-certs
  - hostPath:
      path: /var/lib/etcd
      type: DirectoryOrCreate
    name: etcd-data
  status: {}
```

```
livenessProbe:
  failureThreshold: 8
  httpGet:
    host: 127.0.0.1
    path: /livez
    port: probe-port
    scheme: HTTP
  initialDelaySeconds: 10
  periodSeconds: 10
  timeoutSeconds: 15
name: etcd
ports:
- containerPort: 2381
  name: probe-port
  protocol: TCP
readinessProbe:
  failureThreshold: 3
  httpGet:
    host: 127.0.0.1
    path: /readyz
    port: probe-port
    scheme: HTTP
  periodSeconds: 1
  timeoutSeconds: 15
resources:
  requests:
    cpu: 25m
    memory: 100Mi
startupProbe:
  failureThreshold: 24
  httpGet:
    host: 127.0.0.1
    path: /readyz
    port: probe-port
    scheme: HTTP
  initialDelaySeconds: 10
  periodSeconds: 10
  timeoutSeconds: 15
volumeMounts:
- mountPath: /var/lib/etcd
  name: etcd-data
- mountPath: /etc/kubernetes/pki/etcd
  name: etcd-certs
hostNetwork: true
priority: 2000001000
priorityClassName: system-node-critical
securityContext:
  seccompProfile:
    type: RuntimeDefault
volumes:
- hostPath:
    path: /etc/kubernetes/pki/etcd
    type: DirectoryOrCreate
  name: etcd-certs
- hostPath:
    path: /var/lib/etcd
    type: DirectoryOrCreate
  name: etcd-data
status: {}
root@controlplane:~# []
```

02 - etcd

etcd đang chạy dưới dạng 1 process với PID 1795

etcd chạy dưới dạng static pod - quản lý bởi kubelet

etcd listen trên 3 ports

- 2379: Kết nối clients như kube-apiserver
- 2380: Kết nối peer clients như các etcd khác
- 2381: Kết nối health metric - chỉ dùng HTTP

etcd dùng các certificate file sau

- Server cert: /etc/kubernetes/pki/etcd/server.crt
- Peer cert: /etc/kubernetes/pki/etcd/peer.crt
- CA cert: /etc/kubernetes/pki/etcd/ca.crt

2- kube-apiserver

```
root@controlplane:~# ps -ef | grep kube-apiserver
root      1758   1597  2 11:37 ?        00:02:04 kube-apiserver --advertise-address=172.30.1.2 --allow-privileged=true --authorization-mode=Node,RBAC --client-ca-file=/etc/kubernetes/pki/ca.crt --enable-admission-plugins=NodeRestriction --enable-bootstrap-token-auth=true --etcd-cafile=/etc/kubernetes/pki/etcd/ca.crt --etcd-certfile=/etc/kubernetes/pki/apiserver-etcd-client.crt --etcd-keyfile=/etc/kubernetes/pki/apiserver-etcd-client.key --etcd-servers=https://127.0.0.1:2379 --kubelet-client-certificate=/etc/kubernetes/pki/apiserver-kubelet-client.crt --kubelet-client-key=/etc/kubernetes/pki/apiserver-kubelet-client.key --kubelet-preferred-address-types=InternalIP,ExternalIP,Hostname --proxy-client-cert-file=/etc/kubernetes/pki/front-proxy-client.crt --proxy-client-key-file=/etc/kubernetes/pki/front-proxy-client.key --requestheader-allowed-names=front-proxy-client --requestheader-client-ca-file=/etc/kubernetes/pki/front-proxy-ca.crt --requestheader-extra-headers-prefix=X-Remote-Extra --requestheader-group-headers=X-Remote-Group --requestheader-username-headers=X-Remote-User --secure-port=6443 --service-account-issuer=https://kubernetes.default.svc.cluster.local --service-account-key-file=/etc/kubernetes/pki/sa.pub --service-account-signing-key-file=/etc/kubernetes/pki/sa.key --service-cluster-ip-range=10.96.0.0/12 --tls-cert-file=/etc/kubernetes/pki/apiserver.crt --tls-private-key-file=/etc/kubernetes/pki/apiserver.key
root      4637   4288  0 12:48 pts/0    00:00:00 psn --color=auto kube-apiserver
```

PID của kube-apiserver: 1758

API Server chạy trên secure port: --secure-port=6443

etcd endpoint: --etcd-servers=https://127.0.0.1:2379

Authorization mode: Node, RBAC

TLS certificate file nằm ở: --tls-cert-file=/etc/kubernetes/pki/apiserver.crt

```

root@controlplane:~$ cat /etc/kubernetes/manifests/kube-apiserver.yaml
apiVersion: v1
kind: Pod
metadata:
  annotations:
    kubeadm.kubernetes.io/kube-apiserver.advertise-address.endpoint: 172.30.1.2:6443
  labels:
    component: kube-apiserver
    tier: control-plane
  name: kube-apiserver
  namespace: kube-system
spec:
  containers:
  - command:
    - kube-apiserver
    - --advertise-address=172.30.1.2
    - --allow-privileged=true
    - --authorization-mode=Node,RBAC
    - --client-ca-file=/etc/kubernetes/pki/ca.crt
    - --enable-admission-plugins=NodeRestriction
    - --enable-bootstrap-token-auth=true
    - --etcd-cafile=/etc/kubernetes/pki/etcd/ca.crt
    - --etcd-certfile=/etc/kubernetes/pki/apiserver-etcd-client.crt
    - --etcd-keyfile=/etc/kubernetes/pki/apiserver-etcd-client.key
    - --etcd-servers=https://127.0.0.1:2379
    - --kubelet-client-certificate=/etc/kubernetes/pki/apiserver-kubelet-client.crt
    - --kubelet-client-key=/etc/kubernetes/pki/apiserver-kubelet-client.key
    - --kubelet-preferred-address-types=InternalIP,ExternalIP,Hostname
    - --proxy-client-cert-file=/etc/kubernetes/pki/front-proxy-client.crt
    - --proxy-client-key-file=/etc/kubernetes/pki/front-proxy-client.key
    - --requestheader-allowed-names=front-proxy-client
    - --requestheader-client-ca-file=/etc/kubernetes/pki/front-proxy-ca.crt
    - --requestheader-extra-headers-prefix=X-Remote-Extra-
    - --requestheader-group-headers=X-Remote-Group
    - --requestheader-username-headers=X-Remote-User
    - --secure-port=6443
    - --service-account-issuer=https://kubernetes.default.svc.cluster.local
    - --service-account-key-file=/etc/kubernetes/pki/sa.pub
    - --service-account-signing-key-file=/etc/kubernetes/pki/sa.key
    - --service-cluster-ip-range=10.96.0.0/12
    - --tls-cert-file=/etc/kubernetes/pki/apiserver.crt
    - --tls-private-key-file=/etc/kubernetes/pki/apiserver.key
    image: registry.k8s.io/kube-apiserver:v1.35.1
    imagePullPolicy: IfNotPresent
    livenessProbe:
      failureThreshold: 8

```

kube-apiserver có phải Static Pod: Có

kube-apiserver chạy trong namespace: kube-system

kubelet dùng file nào để tạo kube-apiserver Pod: /etc/kubernetes/manifests/kube-apiserver.yaml

Option --etcd-servers có giá trị: Kết nối nội vi với etcd

3- kube-controller-manager

PID: 4699

Secure port: Không có

etcd endpoint: Không có - chỉ có thể kết nối với kube-apiserver

Authorization mode: Không có - chỉ có config tại

--authentication-kubeconfig=/etc/kubernetes/controller-manager.conf

--authorization-kubeconfig=/etc/kubernetes/controller-manager.conf

TLS certificate file:

--client-ca-file=/etc/kubernetes/pki/ca.crt

--cluster-signing-cert-file=/etc/kubernetes/pki/ca.crt

--requestheader-client-ca-file=/etc/kubernetes/pki/front-proxy-ca.crt

--root-ca-file=/etc/kubernetes/pki/ca.crt

Static Pod: Có

Namespace: kube-system

Manifest file: /etc/kubernetes/manifests/kube-controller-manager.yaml

```
K L L K C O D A Exam Desktop Editor Tab 1 +
root@controlplane:~# ps -ef | grep kube-controller-manager
root      1799      1638      1 12:49 ?        00:00:14 kube-controller-manager --authentication-kubeconfig=/etc/kubernetes/controller-manager.conf --authorization-kubeconfig=/etc/kubernetes/controller-manager.conf --bind-address=127.0.0.1
--client-ca-file=/etc/kubernetes/pki/ca.crt --cluster-name=kubernetes --cluster-signing-cert-file=/etc/kubernetes/pki/ca.crt --cluster-signing-key-file=/etc/kubernetes/pki/ca.key --controllers=*,bootstrapsigner,tokencleaner --kubeconfig
/etc/kubernetes/controller-manager.conf --leader-elect=true --requestheader-client-ca-file=/etc/kubernetes/pki/front-proxy-ca.crt --root-ca-file=/etc/kubernetes/pki/ca.crt --service-account-private-key-file=/etc/kubernetes/pki/sa.key -
service-cluster-ip-range=10.96.0.0/12 --use-service-account-credentials=true --allocate-node-cidrs=true --cluster-cidr=192.168.0.0/16
root      3986      3960      0 13:09 pts/0    00:00:00 grep --color=auto kube-controller-manager
root@controlplane:~# cat /etc/kubernetes/manifests/kube-controller-manager.yaml
apiVersion: v1
kind: Pod
metadata:
  labels:
    component: kube-controller-manager
    tier: control-plane
  name: kube-controller-manager
  namespace: kube-system
spec:
  containers:
    - command:
        - kube-controller-manager
        - --authentication-kubeconfig=/etc/kubernetes/controller-manager.conf
        - --authorization-kubeconfig=/etc/kubernetes/controller-manager.conf
        - --bind-address=127.0.0.1
        - --client-ca-file=/etc/kubernetes/pki/ca.crt
        - --cluster-name=kubernetes
        - --cluster-signing-cert-file=/etc/kubernetes/pki/ca.crt
        - --cluster-signing-key-file=/etc/kubernetes/pki/ca.key
        - --controllers=*,bootstrapsigner,tokencleaner
        - --kubeconfig=/etc/kubernetes/controller-manager.conf
        - --leader-elect=true
        - --requestheader-client-ca-file=/etc/kubernetes/pki/front-proxy-ca.crt
        - --root-ca-file=/etc/kubernetes/pki/ca.crt
        - --service-account-private-key-file=/etc/kubernetes/pki/sa.key
        - --service-cluster-ip-range=10.96.0.0/12
        - --use-service-account-credentials=true
        - --allocate-node-cidrs=true
        - --cluster-cidr=192.168.0.0/16
      image: registry.k8s.io/kube-controller-manager:v1.35.1
      imagePullPolicy: IfNotPresent
      livenessProbe:
        failureThreshold: 8
        httpGet:
          host: 127.0.0.1
          path: /healthz
          port: probe-port
          scheme: HTTPS
      initialDelaySeconds: 10
```

04- kube-scheduler

```
K L L K C O D A ExamDesktop Editor Tab1 + 32m
root@controlplane:~# ps -ef | grep kube-scheduler
root      1818      1624  0 12:49 ?        00:00:16 kube-scheduler --authentication-kubeconfig=/etc/kubernetes/scheduler.conf --authorization-kubeconfig=/etc/kubernetes/scheduler.conf --bind-address=127.0.0.1 --kubeconfig=/etc/kubernet
s/scheduler.conf --leader-elect=true
root      4265      3969  0 13:34 pts/0    00:00:00 grep --color=auto kube-scheduler
root@controlplane:~# cat /etc/kubernetes/manifests/kube-scheduler.yaml
apiVersion: v1
kind: Pod
metadata:
  labels:
    component: kube-scheduler
    tier: control-plane
  name: kube-scheduler
  namespace: kube-system
spec:
  containers:
  - command:
    - kube-scheduler
    - --authentication-kubeconfig=/etc/kubernetes/scheduler.conf
    - --authorization-kubeconfig=/etc/kubernetes/scheduler.conf
    - --bind-address=127.0.0.1
    - --kubeconfig=/etc/kubernetes/scheduler.conf
    - --leader-elect=true
    image: registry.k8s.io/kube-scheduler:v1.35.1
    imagePullPolicy: IfNotPresent
    livenessProbe:
      failureThreshold: 8
      httpGet:
        host: 127.0.0.1
        path: /livez
        port: probe-port
        scheme: HTTPS
      initialDelaySeconds: 10
      periodSeconds: 10
      timeoutSeconds: 15
    name: kube-scheduler
    ports:
    - containerPort: 10259
      name: probe-port
      protocol: TCP
    readinessProbe:
      failureThreshold: 3
      httpGet:
        host: 127.0.0.1
        path: /readyz
        port: probe-port
        scheme: HTTPS
```

PID: 1818

Chạy trên secure port:

ports:

- containerPort: 10259

name: probe-port

schem: HTTPS

etcd endpoint: ko có

Authorization mode: ko có, chỉ có auth config

TLS certificate file: Ko có

Static Pod: Có

namespace: kube-system

kubelet dùng file nào để tạo Pod:

/etc/kubernetes/manifests/kube-scheduler.yaml

05 – kubelet

```

root@controlplane:~# ps -aux | grep kubelet
root      1581    2.0  3.0 1896388 71216 ?        Ssl  14:22  0:40 /usr/bin/kubelet --bootstrap-kubeconfig=/etc/kubernetes/bootstrap-kubelet.conf --kubeconfig=/etc/kubernetes/kubelet.conf --config=/var/lib/kubelet/config.yaml --container-runtime-endpoint unix:///run/containerd/containerd.sock --cgroup-driver=systemd --eviction-hard imagefs.available<5%,memory.available<100%,nodefs.available<5% --fail-swap-on=false
root      1866    3.5 13.1 1516124 303604 ?        Ssl  14:22  1:08 kube-apiserver --advertise-address=172.30.1.2 --allow-privileged=true --authorization-mode=Node,RBAC --client-ca-file=/etc/kubernetes/pki/ca.crt --enable-admission-plugins=NodeRestriction --enable-bootstrap-token-auth=true --etcd-cafile=/etc/kubernetes/pki/etcd/ca.crt --etcd-certfile=/etc/kubernetes/pki/apiserver-etcd-client.crt --etcd-keyfile=/etc/kubernetes/pki/apiserver-etcd-client.key --etcd-servers=https://127.0.0.1:2379 --kubelet-client-certificate=/etc/kubernetes/pki/apiserver-kubelet-client.crt --kubelet-client-key=/etc/kubernetes/pki/apiserver-kubelet-client.key --kubelet-preferred-address-types=InternalIP,ExternalIP,Hostname --proxy-client-cert-file=/etc/kubernetes/pki/front-proxy-client.crt --proxy-client-key-file=/etc/kubernetes/pki/front-proxy-client.key --requestheader-allowed-names=front-proxy-client --requestheader-client-ca-file=/etc/kubernetes/pki/front-proxy-ca.crt --requestheader-extra-headers-prefix=X-Remote-Extra --requestheader-group=headers-X-Remote-Group --requestheader-username=headers-X-Remote-User --secure-port=6443 --service-account-issuer=https://kubernetes.default.svc.cluster.local --service-account-key-file=/etc/kubernetes/pki/sa.pub --service-account-signing-key-file=/etc/kubernetes/pki/sa.key --service-cluster-ip-range=10.96.0.0/12 --tls-cert-file=/etc/kubernetes/pki/apiserver.crt --tls-private-key-file=/etc/kubernetes/pki/apiserver.key
root      4237    0.0  0.0   3528 1880 pts/0    S+   14:54  0:00 grep --color=auto kubelet

root@controlplane:~# ps -ef | grep -E "PID|kubelet"
UID          PID    PPID  C STIME TTY          TIME CMD
root        1581    1  14:22 ?        00:00:40 /usr/bin/kubelet --bootstrap-kubeconfig=/etc/kubernetes/bootstrap-kubelet.conf --kubeconfig=/etc/kubernetes/kubelet.conf --config=/var/lib/kubelet/config.yaml --container-runtime-endpoint unix:///run/containerd/containerd.sock --cgroup-driver=systemd --eviction-hard imagefs.available<5%,memory.available<100%,nodefs.available<5% --fail-swap-on=false
root        1866    1563  3 14:22 ?        00:01:08 kube-apiserver --advertise-address=172.30.1.2 --allow-privileged=true --authorization-mode=Node,RBAC --client-ca-file=/etc/kubernetes/pki/ca.crt --enable-admission-plugins=NodeRestriction --enable-bootstrap-token-auth=true --etcd-cafile=/etc/kubernetes/pki/etcd/ca.crt --etcd-certfile=/etc/kubernetes/pki/apiserver-etcd-client.crt --etcd-keyfile=/etc/kubernetes/pki/apiserver-etcd-client.key --etcd-servers=https://127.0.0.1:2379 --kubelet-client-certificate=/etc/kubernetes/pki/apiserver-kubelet-client.crt --kubelet-client-key=/etc/kubernetes/pki/apiserver-kubelet-client.key --kubelet-preferred-address-types=InternalIP,ExternalIP,Hostname --proxy-client-cert-file=/etc/kubernetes/pki/front-proxy-client.crt --proxy-client-key-file=/etc/kubernetes/pki/front-proxy-client.key --requestheader-allowed-names=front-proxy-client --requestheader-client-ca-file=/etc/kubernetes/pki/front-proxy-ca.crt --requestheader-extra-headers-prefix=X-Remote-Extra --requestheader-group=headers-X-Remote-Group --requestheader-username=headers-X-Remote-User --secure-port=6443 --service-account-issuer=https://kubernetes.default.svc.cluster.local --service-account-key-file=/etc/kubernetes/pki/sa.pub --service-account-signing-key-file=/etc/kubernetes/pki/sa.key --service-cluster-ip-range=10.96.0.0/12 --tls-cert-file=/etc/kubernetes/pki/apiserver.crt --tls-private-key-file=/etc/kubernetes/pki/apiserver.key
root        4239    4010  0 14:54 pts/0    00:00:00 grep --color=auto -E PID|kubelet

root@controlplane:~# systemctl status kubelet
● kubelet.service - kubelet: The Kubernetes Node Agent
   Loaded: loaded (/usr/lib/systemd/system/kubelet.service; enabled; preset: enabled)
   Drop-In: /usr/lib/systemd/system/kubelet.service.d
            └─10-kubeadm.conf
   Active: active (running) since Fri 2026-05-08 14:22:48 UTC; 32min ago
     Docs: https://kubernetes.io/docs/
   Main PID: 1581 (kubelet)
    Tasks: 11 (limit: 2649)
   Memory: 69.3M (peak: 69.9M)
     CPU: 40.437s
   CGroup: /system.slice/kubelet.service
            └─1581 /usr/bin/kubelet --bootstrap-kubeconfig=/etc/kubernetes/bootstrap-kubelet.conf --kubeconfig=/etc/kubernetes/kubelet.conf --config=/var/lib/kubelet/config.yaml --container-runti

```

kubelet có đang chạy

kubelet chạy dưới dạng service

PID của kubelet: 1581

kubelet chạy dưới user: root

```
root@controlplane:~$ crictl ps | grep nginx
root@controlplane:~$ kubectl get pods -A -o wide | grep nginx
root@controlplane:~$ kubectl get pods -o wide
No resources found in default namespace.
```

Pod nginx chạy trên node nào? - Đường như không tồn tại

Kubelet hay scheduler tạo container? - Kubelet tương tác với CRI để tạo container

Container runtime có đang chạy nginx container không? - Dường như là không

kubelet có giao tiếp trực tiếp với etcd không? Không